

1 Introduction

Social media has become a major source of information for many Americans. Leading up to the 2016 US presidential election, around 14% of Americans indicated social media as their primary source of news (Allcott and Gentzkow (2017)), and by 2019, over 70% of Americans reported receiving at least *some* of their news from social media (Levy (2021)). At the same time, there is growing concern about viral misinformation in social media, including made-up news stories such as those claiming that there were no mass shootings under Donald Trump or that Hillary Clinton approved ISIS weapon sales.¹ Some recent evidence also suggests that misinformation on social media has impacted critical decisions such as vaccinations against COVID-19 (see Pennycook et al. (2018); Pennycook et al. (2020b)).

Although there is yet no consensus on what promotes the spread of falsehoods and misleading content on social media, two sets of factors have been emphasized. The first is the presence of echo chambers, which arise when individuals communicate and share content with like-minded users (Sunstein (2018), Lazer et al. (2018)). Törnberg (2018) and Vicario et al. (2016) show that echo chambers reinforce existing political viewpoints and tend to propagate misinformation. Social media, much more than traditional media, allows individual users to choose who and what they listen to, and thus echo chambers may be an unavoidable side effect. However, there is also evidence that echo chambers are a result of the “filter bubbles” that platform algorithms create (see Levy (2021) on Facebook). The second factor conjectured to have fueled misinformation is the general political polarization in many countries, and especially the United States,² and there is some preliminary evidence suggesting that polarization has contributed to selective exposure to questionable content on social media as well (Guess et al. (2018)). Despite the importance and salience of these issues, we do not currently have a framework to understand how online interactions impact the spread of misinformation and what factors shape incentives for sharing low-reliability content.

In this paper, we develop a parsimonious model of online sharing behavior in the presence of misinformation, and as a first step, we focus on the behavior of fully Bayesian agents.³ Our model is inhabited by a set of N agents, each of which has a prior about the state of the world (“ideological bias”), and is connected to the rest of the users via a network, which is given by agents’ friends and acquaintances, and is also shaped by the algorithms of the social media platform. A news article, defined by an underlying type (truthful or containing misinformation), a message (right-wing or left-wing), and a level of reliability (which determines the likelihood of misinformation), is then seeded at one of the agents. The message and the level of reliability of the article are common knowledge, while whether it is truthful or contains misinformation is unobserved, and agents form beliefs about this

¹See <https://www.snopes.com/fact-check/mass-shootings-under-trump/> and <https://www.cnn.com/2016/12/30/read-all-about-it-the-biggest-fake-news-stories-of-2016.html>, respectively.

²While there has been some debate about whether polarization has been mainly among politicians (see Fiorina et al. (2008) and Prior (2013)), there is considerable evidence that polarization has also risen among the general public (see Pew Research Center (2014) and Abramowitz (2010)).

³Myopic reactions and biased behavior appear to play some role, for example, via the “confirmation bias” in social media behavior (see, e.g., Buchanan (2020) and Pennycook and Rand (2019b)), but we believe that the Bayesian benchmark we construct already generates a number of empirically-relevant and rich results. We view incorporating realistic and relevant behavioral biases as a next step in this research agenda.

component.

Given these beliefs, the agent in question decides whether to ignore, dislike, or share the news article. If it is shared, the article moves from the agent sequentially to her connections on social media, who are then faced with the same choices. If the article is ignored or disliked, it does not get past the agent. We assume that agents receive utility when their shared content is re-shared and incur a cost when it is disliked. The former aspect captures the role of positive engagement in social media, while the latter represents the reputation loss from being called out for sharing content containing misinformation. Agents additionally receive utility from disliking (or calling out) items that they believe contain misinformation.

We characterize the Bayesian-Nash equilibria of this sequential game and prove that these equilibria always exist and are in cutoff strategies. In particular, our payoff structure implies that an individual will share any item that she believes is truthful with a high probability and will dislike articles that she believes to contain falsehoods. Items with intermediate beliefs will be ignored. Beliefs about the truthfulness of articles are formed on the basis of the article’s reliability and message, and agents’ prior beliefs/ideology. Moreover, we establish that ours is a game of strategic complements: when others are more likely to share an item, each agent also becomes more likely to do so. As a result, we show that the set of equilibria forms a lattice, with well-defined most-sharing and least-sharing equilibria. We present a number of comparative statics that describe how these extremal equilibria shift. For example, low-reliability articles are shared less, while articles that are more sensational are shared more. As expected, greater reputation costs reduce sharing uniformly.

We present two main results. First, we study the implications of the (social media) network structure. We establish non-monotone comparative statics with respect to the degree of homophily (which determines how likely agents are to be connected to others who are ideologically similar to them). Low levels of homophily ensure that agents are likely to be exposed to cross-cutting content, including “counter-attitudinal articles” that advocate views opposed to theirs. This in turn ensures that misinformation is unlikely to survive for very long. Perhaps paradoxically, for high-reliability articles, an increase in homophily reduces content virality. This is because greater homophily makes it less likely that an article escapes a given community, reducing its circulation throughout the network.

More interestingly and ominously, when relevant news items have low reliability, homophily increases virality. This is because, countering the circulation effect, high homophily also creates a perverse incentive effect: knowing that shared articles will be seen by like-minded individuals, agents become more likely to share questionable content. Strategic complementarities amplify this effect, because when others are expected to share, the benefits from sharing and reduces the likelihood of being called out for spreading misinformation are greater. It is particularly telling that homophily leads to the viral spread of low-reliability content, which are the ones more likely to contain misinformation.

We also show that political polarization and politically divisive articles are more likely to spread virally when they are low-reliability and the level of homophily is already high, generating an echo chamber-like social media environment. Strategic complementarities tend to amplify these pernicious effects of political polarization and divisive content as well.

Our second main result turns to social media platforms’ algorithm design choices and establishes

a striking result: when the relevant articles have low reliability, social media platforms that aim to maximize engagement will tend to design algorithms that increase homophily and create filter bubbles, propagating misinformation. Intuitively, high-reliability content tends to spread anyway because most users recognize it as such and share it, and low homophily contributes to its spread by increasing its circulation throughout the network. In contrast, low-reliability content will be ignored or disliked by agents who disagree with its message and believe it to contain misinformation. Engagement with low-reliability content can be increased if the platform ensures that it remains among users ideologically aligned with its message, who would be willing to share it with like-minded others without fear of being called out by users with different ideologies. Hence, creating filter bubbles becomes an attractive strategy for engagement-maximizing platforms. It is particularly troublesome that such filter bubbles are created precisely when the relevant content is low-reliability and likely to contain misinformation.

If platform algorithms are propagating misinformation, can public policy counter and discourage this type of behavior?⁴ The answer is yes, but with some caveats. In the last part of the paper, we discuss four different types of regulatory policies, and in each case, we show how they may reduce misinformation but also point out the possibility that, if they are not designed well, they can backfire and exacerbate the problem.

First, we discuss regulations that force platforms to reveal the provenance of articles, making it easier for users to identify falsehoods (e.g., claims originating from less reputable sources, such as InfoWars). Though generally useful, such provenance information can backfire when it instills a sense of “false security” that the content has been verified by previous social media users. Second, we look at potential censorship of articles identified by a regulator as likely containing misinformation. While censorship can help reduce the viral spread of misinformation, it also generates an “implied truth” effect (Pennycook et al. (2020a)) that contributes to the viral spread of questionable content that escapes censorship. Third, we discuss “performance targets”, where the regulator places limits on the amount of misinformation that circulates on the platform. Such targets tend to better align user and regulator preferences, but unless they appropriately target sensational content, platforms can meet the required performance limits by focusing on less sensational material that was unlikely to become viral in any case, and the least reliable content may still continue to circulate virally. Lastly, we show how direct regulation of platform algorithms can reduce misinformation, but also point out that the non-monotone effects of homophily imply that such regulations need to be finely calibrated.

Related Literature. Our paper builds on a large body of work on models of misinformation. In addition to the literature mentioned previously, several other papers in this literature are related to our findings.

Much previous work has focused on the susceptibility of boundedly-rational agents to engage with misinformation. In Acemoglu et al. (2010) and Acemoglu et al. (2013), the existence of persuasive agents can impede information aggregation and enable misinformed beliefs to survive, and sometimes even become dominant, in the population. In Mostagir et al. (2021) and Mostagir and Siderius (2021),

⁴As of August 2021, federal law protects social media platforms from being held responsible for content posted by its users (see Section 230 of the Communications Decency Act of 1996, discussed by <https://hbr.org/2021/08/its-time-to-update-section-230>).

a strategic principal who wants to persuade agents of an incorrect belief can distort the learning process by leveraging social connections and echo chambers to propagate misinformation. Similarly, models of misinformation “contagion”— without Bayesian agents or strategic decisions — have been studied in [Budak et al. \(2011\)](#), [Nguyen et al. \(2012\)](#), and [Törnberg \(2018\)](#). Our contribution relative to this literature is the possibility that misinformation spreads because of the strategic interactions of Bayesian agents and is exacerbated by profit-maximizing platform algorithms.

There is a growing literature on information design by platforms, building for the most part on the concept of Bayesian persuasion ([Kamenica and Gentzkow \(2011\)](#) and [Kamenica \(2019\)](#)). [Candogan and Drakopoulos \(2020\)](#) study how a platform with private knowledge of content’s accuracy should optimally signal to rational users whether to engage with it, while [Chen and Papanastasiou \(2021\)](#) and [Keppo et al. \(2019\)](#) consider more manipulative actions by platforms, including strategic seeding of information or “cheap talk” signals about quality. Also related are works on reputation and media bias. Motivated by the 2016 presidential election, [Allcott and Gentzkow \(2017\)](#) study the incentives of certain outlets to present misleading news, while [Gentzkow and Shapiro \(2006\)](#), [Hsu et al. \(2020\)](#) and [Allon et al. \(2021\)](#) explore other strategic reasons for media bias. Our paper contributes to this literature by highlighting the role of ideological leaning, strategic interactions, ideological homophily, and platform algorithms.

The most closely related work to ours is [Papanastasiou \(2020\)](#), who studies a model where agents hold heterogenous ideological beliefs and digest (and potentially share) a news article sequentially. Our work is different in three important dimensions. First, [Papanastasiou \(2020\)](#) focuses on costly inspection, which makes sharing decisions strategic substitutes, while our model generates strategic complementarities, because individuals care about further shares of the content they share.⁵ All of our results and formal analysis turn on strategic complementarities. Second, and relatedly, echo chambers play no role in [Papanastasiou \(2020\)](#).⁶ Third, our analysis of engagement-maximization by the platform and its implications for the spread of low-reliability content has no counterpart in [Papanastasiou \(2020\)](#) or any other work in this area we are aware of.⁷

The rest of the paper is organized as follows. The next section introduces our basic environment and describes the information structure and payoffs. Section 3 characterizes the (Bayesian-Nash) equilibria of this model and provides some basic comparative static results. Section 4 studies the effects of homophily by focusing on a special class of sharing networks that correspond to a set of “islands” of like-minded individuals who are less closely linked to those in other islands. Section 5 endogenizes the sharing network as a result of the algorithmic choices of the platform that aims to maximize engagement. Section 6 discusses a range of regulations aimed at containing misinformation.

⁵Our reading of the evidence is that strategic complementarities are more relevant for social media behavior than strategic substitutabilities. For example, [Eckles et al. \(2016\)](#) find evidence that feedback or “encouragement” from peers about Facebook posts have contributed significantly to future behavior and posting. See also [Taylor and Eckles \(2018\)](#) and [Aral and Dhillon \(2018\)](#).

⁶As already noted, echo chambers appear central to the spread of misinformation in practice. See, for example, [Lee et al. \(2011\)](#), [Törnberg \(2018\)](#), [Centola \(2010\)](#), and [Centola and Macy \(2007\)](#)

⁷[Papanastasiou \(2020\)](#) also discusses platform incentives, but assumes that the platform is interested in limiting misinformation. Our reading of the evidence in this instance, too, favors our interpretation, where platforms such as Facebook are (or at the very least used to be before regulatory pressure mounted) fairly indifferent to the presence of misinformation but strongly prioritize engagement maximization.